Synthetic Paper A: Evidence-Grounded Generation for Scientific Q&A;

Fixture ID SYN-A - Version 1.0 - Intended for project-level retrieval and paper comparison

TEST DATA DISCLOSURE: This paper is fictional. All study descriptions, sample sizes, measurements, and results below are invented solely to test MAIROS workflows. Do not cite as published research.

Abstract

This fictional study tests whether retrieving source passages before generation improves evidence support in scientific question answering. It compares a parametric-only answerer with a retrieval-augmented answerer on a small constructed question set. The purpose here is to supply stable, internally consistent test passages for upload, retrieval, comparison, gap analysis, and evaluation workflows.

Research question and hypothesis

RQ-A: On questions whose answers appear in a supplied evidence collection, does retrieval-augmented generation increase the proportion of factual claims supported by a cited passage compared with a parametric-only baseline? H-A: retrieval-augmented generation will increase supported-claim rate. This is a fictional, testable hypothesis, not an established result.

Methods

The fixture describes 100 synthetic scientific questions with one short evidence passage per question. Two answer conditions are compared: (1) Parametric-only, with no passage supplied at answer time; and (2) Retrieve-then-generate, with the top five passages from the fixture collection available to the generator. Two human-review-style labels are represented as invented counts: supported answer means the central factual claim is entailed by a retrieved passage; citation entailment means the cited passage directly supports the claim.

Dataset and controls

All 100 question/evidence pairs are invented and cover general methods concepts. The same questions and answer rubric are intended for both conditions. The fixture does not contain real participant data, a real benchmark, or a real model run. It does not establish statistical significance or generalization to real scientific literature.

Synthetic Paper A: Evidence-Grounded Generation for Scientific Q&A; — results and interpretation

TEST DATA DISCLOSURE: Every value on this page is invented illustrative data, not a real empirical finding.

Measure (invented)	Parametric-only	Retrieve-then-generate
Central claim supported	62 / 100 (62%)	81 / 100 (81%)
Citation entails claim	Not applicable: no citations	74 / 100 (74%)
Unsupported central claim	38 / 100 (38%)	19 / 100 (19%)

Illustrative interpretation

Within this fictional fixture, the retrieval condition has a higher supported-claim count than the parametric-only condition. The difference is a convenient positive signal for checking whether MAIROS retrieves passages about evidence grounding and whether the answer flow can cite them. It must not be reported as a real research result.

Limitations and open questions

The constructed questions are small and synthetic; no real corpus, model, annotator agreement, confidence interval, or cost measurement is provided. Citation entailment is lower than supported-answer rate in the invented counts, leaving a test gap: can citation checking identify unsupported citations, especially when a relevant passage is paraphrased? A follow-up test could vary retrieval depth and paraphrase distance while holding questions fixed.

Comparison hooks

Compare SYN-A with SYN-B on the role of graph links and retrieval quality, and with SYN-C on document segmentation. Distinguish each paper’s research question, intervention, data, metric, limitation, and evidence. The values in all three fixtures are invented, so compare them as application test cases only.